

MYSQL TRIGGERS

Create database:

```
1 • CREATE DATABASE trigger_demo;  
2 USE trigger_demo;
```

```
✓ 1 22:36:37 CREATE DATABASE trigger_demo  
✓ 2 22:36:37 USE trigger_demo
```

Step 1: Create the main table

```
3 • CREATE TABLE employees(  
4     emp_id INT PRIMARY KEY,  
5     emp_name VARCHAR(50),  
6     dept_id INT,  
7     salary DECIMAL(10,2)  
8 );
```

Output				
Action Output				
#	Time	Action	Message	
1	22:36:37	CREATE DATABASE trigger_demo	1 row(s) affected	
2	22:36:37	USE trigger_demo	0 row(s) affected	
3	22:38:20	CREATE TABLE employees(emp_id INT PRIMARY KEY, emp_name VARCHAR...	0 row(s) affected	

Step 2: Create the audit table

```
9 • CREATE TABLE employee_audit(  
10     audit_id INT AUTO_INCREMENT PRIMARY KEY,  
11     emp_id INT,  
12     emp_name VARCHAR(50),  
13     action_type VARCHAR(20),  
14     action_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
15 );
```

```
✓ 4 22:39:46 CREATE TABLE employee_audit( audit_id INT AUTO_INCREMENT PRIMARY KE... 0 row(s) affected
```

Step 3: Trigger 1 — AFTER INSERT

5 22:40:41 CREATE TRIGGER after_employee_insert AFTER INSERT ON employees FOR EACH ROW 0 row(s) affected

```
27 • INSERT INTO employees
28   VALUES (106, 'Arjun', 1, 52000);
29
30 • SELECT * FROM employee_audit;
```

Result Grid

	audit_id	emp_id	emp_name	action_type	action_time
▶	1	106	Arjun	INSERT	2026-08-11 22:41:22
*	NULL	NULL	NULL	NULL	NULL

6 22:41:22 INSERT INTO employees VALUES (106, 'Arjun', 1, 52000) 1 row(s) affected

7 22:41:22 SELECT * FROM employee_audit LIMIT 0, 1000 1 row(s) returned

Step 4: Trigger 2 — AFTER DELETE

8 22:42:45 CREATE TRIGGER after_employee_delete AFTER DELETE ON employees FOR EACH ROW 0 row(s) affected

```
43 • DELETE FROM employees
44   WHERE emp_id = 106;
45
46 • SELECT * FROM employee_audit;
```

Result Grid

	audit_id	emp_id	emp_name	action_type	action_time
▶	1	106	Arjun	INSERT	2026-08-11 22:41:22
	2	106	Arjun	DELETE	2026-08-11 22:43:13
*	NULL	NULL	NULL	NULL	NULL

9 22:43:13 DELETE FROM employees WHERE emp_id = 106 1 row(s) affected

10 22:43:13 SELECT * FROM employee_audit LIMIT 0, 1000 2 row(s) returned

Step 5: Trigger 3 — AFTER UPDATE

11 22:44:35 CREATE TRIGGER after_employee_update AFTER UPDATE ON employees FOR EACH ROW 0 row(s) affected

Step 6: Trigger 4 — BEFORE INSERT

12 22:45:33 CREATE TRIGGER check_salary BEFORE INSERT ON employees FOR EACH ROW 0 row(s) affected

```

70 • INSERT INTO employees
71 VALUES (107, 'Ravi', 1, 15000);

```

Context Help Snippets

Output

Action Output

#	Time	Action	Message
✓ 11	22:44:35	CREATE TRIGGER after_employee_update AFTER UPDATE ON employees FOR ...	0 row(s) affected
✓ 12	22:45:33	CREATE TRIGGER check_salary BEFORE INSERT ON employees FOR EACH R...	0 row(s) affected
✗ 13	22:46:02	INSERT INTO employees VALUES (107, 'Ravi', 1, 15000)	Error Code: 1644. Salary cannot be less than 20000

INSERT INTO employees VALUES (108, 'Meena', 1, 30000);

✗ 13	22:46:02	INSERT INTO employees VALUES (107, 'Ravi', 1, 15000)	Error Code: 1644. Salary cannot be less than 20000
✓ 14	22:47:05	INSERT INTO employees VALUES (108, 'Meena', 1, 30000)	1 row(s) affected

Step 7: Trigger 5 — BEFORE UPDATE

```

73 DELIMITER $$
74 • CREATE TRIGGER prevent_salary_reduction
75 BEFORE UPDATE ON employees
76 FOR EACH ROW
77 BEGIN
78     IF NEW.salary < OLD.salary THEN
79         SIGNAL SQLSTATE '45000'
80         SET MESSAGE_TEXT = 'Salary reduction is not allowed';
81     END IF;
82 END $$
83 DELIMITER ;

```

Context Help Snippets

Output

Action Output

#	Time	Action	Message
✗ 13	22:46:02	INSERT INTO employees VALUES (107, 'Ravi', 1, 15000)	Error Code: 1644. Salary cannot be less than 20000
✓ 14	22:47:05	INSERT INTO employees VALUES (108, 'Meena', 1, 30000)	1 row(s) affected
✓ 15	22:48:09	CREATE TRIGGER prevent_salary_reduction BEFORE UPDATE ON employees F...	0 row(s) affected

Step 8: Add employee 101 so you can test updates

```

84 • INSERT INTO employees VALUES (101, 'TestUser', 1, 50000);

```

Output

Action Output

#	Time	Action	Message
✓ 14	22:47:05	INSERT INTO employees VALUES (108, 'Meena', 1, 30000)	1 row(s) affected
✓ 15	22:48:09	CREATE TRIGGER prevent_salary_reduction BEFORE UPDATE ON employees F...	0 row(s) affected
✓ 16	22:48:57	INSERT INTO employees VALUES (101, 'TestUser', 1, 50000)	1 row(s) affected

```

85 • UPDATE employees
86   SET salary = 40000
87   WHERE emp_id = 101;
88 • UPDATE employees
89   SET salary = 60000
90   WHERE emp_id = 101;
91
92 • SELECT * FROM employee_audit;

```

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Result Grid				
Filter Rows:				
Edit:    				
Export/Import:  				
Wrap C				
audit_id	emp_id	emp_name	action_type	action_time
1	106	Arjun	INSERT	2026-08-11 22:41:22
2	106	Arjun	DELETE	2026-08-11 22:43:13
3	108	Meena	INSERT	2026-08-11 22:47:05
4	101	TestUser	INSERT	2026-08-11 22:48:57
5	101	TestUser	UPDATE	2026-08-11 22:49:53
* NULL	NULL	NULL	NULL	NULL

Result Grid

Form Editor

employee_audit 55 x

Apply

Revert

Context Help

Snippets

Output

Action Output

#	Time	Action	Message
17	22:49:35	UPDATE employees SET salary = 40000 WHERE emp_id = 101	Error Code: 1644. Salary reduction is not allowed
18	22:49:53	UPDATE employees SET salary = 60000 WHERE emp_id = 101	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
19	22:49:53	SELECT * FROM employee_audit LIMIT 0, 1000	5 row(s) returned